

Linking Alkali Distribution and Mn Redox Gradients in Manganese Oxide Nanocrystals: A Combined STEM-EELS and First-Principles Study

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Understanding how cation distribution governs electronic structure and morphology in transition-metal oxides remains a key challenge in the rational design of catalytic materials. In this work, we combine scanning transmission electron microscopy with electron energy-loss spectroscopy (STEM-EELS) and first-principles calculations (density functional theory, DFT) to establish structure-property relationships in two model manganese oxides: tunnel-structured $\text{K}_2\text{Mn}_8\text{O}_{16}$ and layered $\text{Na}_2\text{Mn}_3\text{O}_7$. The materials were synthesized via precipitation, hydrothermal, and solid-state routes, enabling systematic variation of precursors, stoichiometry, and thermal treatment. The STEM-EELS spectrum imaging at the Mn $L_{2,3}$ and O-K edges provides spatially resolved insight into Mn valence states and Mn-O hybridization within individual nanocrystals. Quantitative analysis of white-line ratios, edge onset shifts, and ELNES features reveals pronounced nanoscale redox heterogeneity, including surface-to-core gradients and site-specific variations correlated with alkali ion distribution. Correlative high resolution STEM and EDX mapping show that crystal morphology (nanorods in $\text{K}_2\text{Mn}_8\text{O}_{16}$ and platelets in $\text{Na}_2\text{Mn}_3\text{O}_7$) directly reflects the underlying crystallographic framework and governs cation localization. Complementary DFT calculations, combined with *ab initio* thermodynamics, were used to determine stable surface terminations, morphology, and oxygen vacancy formation energies. For the most stable structural models, computed electronic structure descriptors were directly compared with EELS energy-loss near-edge structures, enabling a consistent interpretation of spectral features in terms of local coordination and oxidation states. By integrating multi-scale spectroscopy with atomistic modeling, this study establishes a mechanistic link between alkali distribution, defect chemistry, and redox behavior in manganese oxide nanocrystals. The results highlight the predictive power of first principles approaches in interpreting experimental spectra and guiding the design of oxide catalysts for thermochemical applications. Acknowledgements: The research is carried out within the OPUS 25 (grant number 2023/49/B/ST4/00740, PI: dr Joanna Gryboś) project, funded by NCN, the Polish National Science Centre.

